Linear Algebra II HW 7

Due Date: on or before 18:00, Apr/17/2025 (Thursday), in tutorial.

1. Let V be an inner product space (not necessary finite dimension) and $T:V\to V$ be a linear transformation such that

$$\langle T(\boldsymbol{u}),T(\boldsymbol{u})\rangle=\langle \boldsymbol{u},\boldsymbol{u}\rangle \text{ for all } \boldsymbol{u}\in V$$

Show that $\langle T(\boldsymbol{u}),T(\boldsymbol{v})\rangle=\langle \boldsymbol{u},\boldsymbol{v}\rangle$ for all $\boldsymbol{u},\boldsymbol{v}\in V$

2. Let V be an inner product space (not necessary finite dimension) and $T:V\to V$ be a mapping such that

$$\langle T(\boldsymbol{u}),\boldsymbol{v}\rangle=\langle \boldsymbol{u},T(\boldsymbol{v})\rangle \text{ for all } \boldsymbol{u},\boldsymbol{v}\in V$$

(a) Show that if $\langle \boldsymbol{u}_1,\boldsymbol{v}\rangle=\langle \boldsymbol{u}_2,\boldsymbol{v}\rangle$ for all $\boldsymbol{v}\in V$, then $\boldsymbol{u}_1=\boldsymbol{u}_2$.

(b) Show that T is a linear transformation.

(c) Show that $\mathrm{Ker}(T)=\mathrm{Im}(T)^{\perp}$ and $\mathrm{Ker}(T)=\mathrm{Ker}(T^2)$.

(c) To show $\mathrm{Ker}(T)\subset \mathrm{Im}(T)^{\perp}$.

3. Let $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ form an orthonormal basis for $\boldsymbol{R}^2$ and let $\boldsymbol{u}$ be a unit vector in $\boldsymbol{R}^2$. If $\boldsymbol{u}\cdot\boldsymbol{u}_1=\frac{1}{2}$, determine the value of $|\boldsymbol{u}\cdot\boldsymbol{u}_2|$.

4. Let $\{\boldsymbol{u}_1,\boldsymbol{u}_2,\boldsymbol{u}_3\}$ be an orthonormal basis for an inner product space V and let

$$\boldsymbol{u}=\boldsymbol{u}_1+2\boldsymbol{u}_2+2\boldsymbol{u}_3 \quad \text{and} \quad \boldsymbol{v}=\boldsymbol{u}_1+7\boldsymbol{u}_3.$$

Determine the value of each of the following:

(a) $\langle \boldsymbol{u},\boldsymbol{v}\rangle$

(b) $\|\boldsymbol{u}\|$ and $\|\boldsymbol{v}\|$

(c) The angle θ between $\boldsymbol{u}$ and $\boldsymbol{v}$

5. The set

$$S = \left\{ \tfrac{1}{\sqrt{2}}, \cos x, \cos 2x, \cos 3x, \cos 4x \right\}$$

is an orthonormal set of vectors in $C[-\pi, \pi]$ with inner product defined by Definition 5.4.3.

(a) Use trigonometric identities to write the function $\sin^4 x$ as a linear combination of elements of *S*.

(b) Use part (a) and Theorem 5.5.7 to find the values of the following integrals:

(i) $\int_{-\pi}^{\pi} \sin^4 x \cos x dx$

(ii) $\int_{-\pi}^{\pi} \sin^4 x \cos 2x dx$

(iii) $\int_{-\pi}^{\pi} \sin^4 x \cos 3x dx$

(iv) $\int_{-\pi}^{\pi} \sin^4 x \cos 4x dx$